

EC466-966: Estimation and Inference in Econometrics

Exercise # 5: Differences-in-Differences

Background: Consider a policy intervention that was implemented in a treatment region in 2020, while a control region did not receive the intervention. You have panel data on unemployment rates for both regions from 2018 to 2022.

Let Y_{ct} denote the unemployment rate in city c at time t , where $c \in \{\text{Treatment, Control}\}$ and $t \in \{2018, 2019, 2020, 2021, 2022\}$. The observed unemployment rates (in percentages) are:

City	2018	2019	2020	2021	2022
Treatment	6.2	6.5	5.8	5.4	5.1
Control	5.8	6.0	6.3	6.5	6.7

1. Calculate the simple difference-in-differences (DiD) estimate of the treatment effect using only the pre-treatment year (2019) and the first post-treatment year (2020). Interpret your result.
2. Suppose we model the unemployment rate as:

$$Y_{ct} = \beta_t + \gamma_c + \delta D_{ct} + u_{ct} \quad (1)$$

where $D_{ct} = 1$ if city c is the treatment city and $t \geq 2020$, and $D_{ct} = 0$ otherwise.

- (a) Explain the meaning of each parameter (β_t , γ_c , δ) in the context of this application.
- (b) What key assumption about the conditional mean function $E(Y_{0ct}|c, t)$ does this specification impose?
- (c) Show algebraically that the DiD estimator identifies δ under this specification.

3. Using potential outcomes notation, let $Y_{ct}(1)$ denote the unemployment rate that would be observed in city c at time t if the treatment were implemented, and $Y_{ct}(0)$ the unemployment rate if the treatment were not implemented.
- (a) Write down the expression for the Average Treatment Effect on the Treated (ATT) at time $t = 2020$.
 - (b) Explain why we cannot directly compute the ATT from observed data alone.
 - (c) State the parallel trends assumption formally using potential outcomes notation.